Typed Anytime-Valid Shift Monitoring under Sparse Labels: Certified Concept Claims and Restarting E-Detectors

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Abstract

A deployed classifier needs different responses to covariate, label and concept shift: keep the model, correct its class prior, or retrain it. We study sequential monitoring that reports which response is warranted, with error control, when only a small random fraction of labels is revealed. Typed Shift Sequential Diagnosis (TSSD) runs three witnesses on one anytime-valid budget: a paired kernel witness for inputs, a gated prevalence witness, and a new posterior-validity witness. The last is a universal-inference e-process whose composite null contains every posterior reachable by a prior and tempering correction of the frozen model, under arbitrary covariate shift and with a label shift at an unknown time. Its crossing is therefore a certified claim that prior correction is insufficient. Every witness also runs as a Shiryaev–Roberts e-detector with average-run-length and finite-window false-alarm bounds. We compare TSSD on identical streams with ten re-implemented detectors and a Bonferroni ensemble of specialists. On controlled streams, TSSD detects all six pure and mixed shift types at a 1.3% null alarm rate and types 180–199 of 200 streams correctly per type. The ensemble types 0 of 194 covariate shifts correctly and makes 149 false concept claims, where TSSD makes none in any experiment. On chronological electricity data, TSSD certifies concept drift in all 260 runs on segments where only a refit repairs the model, and in none where a prior correction suffices. TSSD is slower than loss-based specialists on concept alternatives, and its certified claim needs labels.

Keywords: dataset shift; concept drift; sequential change detection; e-processes; universal inference; label shift; anytime-valid inference

1 Introduction

A deployed classifier can fail after deployment for different reasons, and the reasons call for different responses. If only the input distribution moves while the conditional law of the label stays the same (covariate shift), the model's posterior remains correct and the usual response is to keep the model and keep watching. If the class prevalence moves while the class-conditional input laws stay fixed (label shift), the posterior is repaired by a prior correction that needs no new training (Saerens et al., 2002; Lipton et al., 2018; Alexandari et al., 2020). If the relation between inputs and labels changes in a way that no prior correction repairs (concept shift), the model must be retrained or must abstain. A monitor that only reports that "something changed" leaves this decision to guesswork. A monitor that names a type without any error control can

trigger an unnecessary retraining, or an inappropriate prior correction, with no bound on how often this happens.

Two further constraints make the problem hard in practice. Labels are sparse: only a small random fraction of outcomes is ever observed, so evidence about $P(Y \mid X)$ accumulates slowly, while unlabelled inputs arrive at every step. And monitoring is continuous: the operator looks at the monitor at data-dependent times and acts on its first alarm, so false-alarm control must hold under optional stopping and, for change detection, should not degrade when the change arrives late.

This paper develops *Typed Shift Sequential Diagnosis* (TSSD) for this setting. TSSD runs three witnesses on one false-alarm budget. Two are paired, source-referenced e-processes: an orthogonalised kernel witness for inputs and a gated prevalence witness that also uses unlabelled cases. The third is new: a *posterior-validity* witness, a universal-inference e-process (Wasserman et al., 2020) whose composite null is the set of all target posteriors that a prior and tempering correction of the frozen model can reach, *with a label shift allowed at an unknown time and with no restriction on the inputs*. Its crossing is therefore a certified statement that prior correction is insufficient, the decision that separates retraining from recalibration. Every witness is evaluated in two modes from the same increments: a test-martingale mode with "ever-alarm" control, and a Shiryaev–Roberts e-detector mode (Shin et al., 2024) that restarts implicitly and controls the average run length.

Contributions.

1. *Action-typed formulation with a certified concept claim.* We define nested monitoring hypotheses aligned with the keep/recalibrate/retrain decision. We also construct a posterior-validity e-process whose false "concept" claims are controlled at level $\alpha/3$ under arbitrary covariate shift and under a label shift at an unknown time (Theorem 3). The required change-robust likelihood maximisation is computed exactly from cached suffix suprema. A certified concavity bound makes the denominator valid even if the optimiser stops early.
2. *Change-detection guarantees.* A Shiryaev–Roberts version of all three witnesses gives a typed alarm with $\mathbb{E}[\tau] \geq A/3$ and $\Pr(\tau \leq m) \leq 3m/A$ under the global null (Theorem 2). Empirically, its detection delay does not grow with the change time, whereas the test-martingale delay does.
3. *Exact validity with finite source pools.* Single-use references drawn without replacement make the guarantees exact for real-data pools (Proposition 1).
4. *A broad empirical comparison.* We compare TSSD on identical streams with ten re-implemented detectors that span the literature from DDM and ADWIN to WATCH, and with a Bonferroni ensemble of specialist detectors. The comparison covers six pure and mixed shift types, graded magnitudes, four label budgets, four change times and eleven ablations, and we report both native thresholds and thresholds matched to a common empirical false-alarm rate. The real-data evaluation uses three datasets: Digits under an exact-validity protocol, natural geographic shift in Covertype, and chronological drift in Electricity. For the two natural shifts the shift type is checked against an offline full-label oracle.

What TSSD is not. TSSD is not the fastest detector of every individual shift type. A specialist detector tuned to one alternative (for instance Page–Hinkley on the log loss for concept shift) detects that alternative sooner, and we report this. TSSD's contribution is a calibrated typed state with a guarantee on its most consequential claim. In our experiments, the obvious alternative, an ensemble of specialists typed by which detector fired, assigns the wrong type to most covariate shifts and makes spurious concept claims under mixed covariate–label shift (Section 7.2). TSSD

does not adapt the model, and its distinction between covariate and label shift is heuristic and is evaluated empirically.

2 Related work and positioning

Shift taxonomies and offline identification. Covariate, label and concept shift are standard categories (Quiñonero-Candela et al., 2009; Moreno-Torres et al., 2012), and concept-drift surveys cover adaptation after drift (Gama et al., 2014; Lu et al., 2019). Black-box shift detection (BBSD; Rabanser et al., 2019) applies two-sample tests to classifier outputs. Label-shift estimation and correction use EM (Saerens et al., 2002), confusion-matrix inversion (Lipton et al., 2018), unified moment views (Garg et al., 2020) and bias-corrected calibration (Alexandari et al., 2020). Closest in spirit to typing is the work of Roschewitz et al. (2025), who identify prevalence, covariate and mixed shift in medical imaging from batches. Their method is batch-based and has no sequential error guarantee.

Sequential and anytime-valid monitoring. Classical change detection (CUSUM, Page–Hinkley (Page, 1954), Shiryaev–Roberts) and stream-mining detectors such as DDM (Gama et al., 2004) and ADWIN (Bifet and Gavaldà, 2007) monitor a single error signal. Test martingales and e-processes (Ville, 1939; Howard et al., 2020; Vovk and Wang, 2021; Ramdas et al., 2023; Waudby-Smith and Ramdas, 2024) support inference under optional stopping. E-detectors (Shin et al., 2024) turn them into nonparametric change detectors with ARL control; our SR mode is an instance. On the shift side, Podkopaev and Ramdas (2022) track the risk of a deployed model with labels and detect *harmful* shifts; Amoukou et al. (2024) remove the need for labels through an error estimator; and Zhang et al. (2026) combine sparse true labels with synthetic labels through prediction-powered inference. Sequential two-sample tests by betting detect input shift with kernel witnesses (Shekhar and Ramdas, 2023) or online classifiers (Jang et al., 2022). Conformal test martingales detect departures from exchangeability (Vovk et al., 2021), and Vovk (2020) decomposes them into concept and label components. WATCH (Prinster et al., 2025) weights conformal martingales to tolerate benign covariate shift while flagging harmful shifts. Bar et al. (2024) and M-FISHER (Khan and Syed, 2025) couple martingale detection with test-time adaptation, and Schirmer et al. (2025) monitor risk during test-time adaptation. Finally, Choi (2026) gives an anytime-valid *confirmation* of a prespecified label-shift correction through a predictive likelihood ratio, a special case of the factors used in our posterior-validity witness.

Positioning. Table 1 compares 17 methods on six properties. To our knowledge, no existing monitor combines the following: (a) sparse random labels together with unlabelled inputs; (b) a single anytime-valid budget for the alarm, with both "ever" and ARL control; and (c) a *typed* claim with its own guarantee under the alternatives of the other types. The posterior-validity witness provides (c) for the decision that matters most operationally, retraining versus prior correction. It does so by testing a composite null that contains every covariate shift and a label shift at an unknown time, using universal inference (Wasserman et al., 2020). Unlike WATCH, it does not need a density-ratio estimate to tolerate covariate shift. Unlike risk tracking, it detects posterior changes that leave the error rate unchanged (for example, label permutations at constant accuracy are not prior-correctable). TSSD does *not* attempt adaptation, and its covariate-versus-label distinction carries no guarantee.

Table 1: Positioning against recent and classical work. Seq.: online stopping rule; AV: anytime-valid false-alarm guarantee (E = ever-alarm/Ville, A = ARL); Lab.: uses revealed labels (S = sparse); Typed: separates shift types; T. guar.: guarantee for a typed claim under the other shift types; Adapt.: performs adaptation. A dash means absent.

Method	Year	Seq.	AV	Lab.	Typed	T. guar.	Monitored signal
DDM (Gama et al., 2004) / ADWIN (Bifet and Gavaldà, 2007)	2004/07	✓	–	✓	–	–	error rate
BBSD (Rabanser et al., 2019)	2019	–	–	–	–	–	output distribution
Concept-shift CTM (Vovk, 2020)	2020	✓	E	✓	concept/label	–	conformal p-values
Retrain-or-not CTM (Vovk et al., 2021)	2021	✓	E	✓	–	–	conformal p-values
Risk tracking (Podkopaev and Ramdas, 2022)	2022	✓	E	✓	harmful	–	risk
Seq. classifier 2ST (Jang et al., 2022)	2022	✓	E	–	–	–	input discriminator
Kernel 2ST by betting (Shekhar and Ramdas, 2023)	2023	✓	E	–	–	–	MMD witness
E-detectors (Shin et al., 2024)	2024	✓	A	–	–	–	generic
Classifier-CUSUM (Evans and G'Sell, 2024)	2024	✓	–	–	label	–	label-shift LR
Harmful shift w/o labels (Amoukou et al., 2024)	2024	✓	E	–	harmful	–	estimated error
POEM (Bar et al., 2024)	2024	✓	E	–	–	–	entropy
WATCH (Prinster et al., 2025)	2025	✓	E	✓	cov./harmful	–	weighted conformal
Shift identification (Roschewitz et al., 2025)	2025	–	–	–	prev./cov./mixed	–	batch tests
M-FISHER (Khan and Syed, 2025)	2025	✓	E	–	–	–	non-conformity
TTA risk monitoring (Schirmer et al., 2025)	2025	✓	E	S	–	–	risk under TTA
PPRM (Zhang et al., 2026)	2026	✓	E	S	harmful	–	PPI risk bound
Label-shift confirmation (Choi, 2026)	2026	✓	E	✓	label	–	prespecified correction
TSSD (this paper)	–	✓	E + A	S	cov./label/concept	concept	paired + UI witnesses

3 Problem formulation: action-typed monitoring

Let the source law be P on $\mathcal{X} \times \{1, \dots, K\}$ with class prior π. A probabilistic classifier g with output $p_g(x) \in \Delta^{K-1}$ is fitted on source data and frozen. At step $t = 1, 2, \dots$ the operator receives a target input X_t, a reveal bit $R_t \in \{0, 1\}$ and, when $R_t = 1$, the label Y_t. The monitor also draws one *single-use* source reference (X'_t, Y'_t). $\mathcal{F}_{t-1}$ denotes everything observed before step t together with all quantities computed from it.

The three textbook shift types matter because they call for different actions. Under covariate shift the source posterior remains correct and the model can usually be kept; under label shift the posterior changes only through the class prior and is repaired by a prior correction (Saerens et al., 2002; Lipton et al., 2018; Alexandari et al., 2020); under concept shift no prior correction repairs the posterior and retraining or abstention is required. We therefore define the monitoring targets through a nested family of hypotheses on the conditional law of the target label,

$$H^{\mathrm{glob}}: \ (X_t, Y_t) \sim P \text{ for all } t,$$
$$H^{C}: \ \mathbb{P}(Y_t = \cdot \mid X_t, \mathcal{F}_{t-1}) = \eta_{\theta_t}(\cdot \mid X_t), \quad \theta_t \in \Theta \text{ piecewise constant with at most one change,}$$

where the *prior-correctable family* is

$$\eta_\theta(y \mid x) = \frac{(q_y/\pi_y)\, p_g(y \mid x)^\beta}{\sum_k (q_k/\pi_k)\, p_g(k \mid x)^\beta}, \qquad \theta = (q, \beta),\ q \in \Delta_\circ^{K-1},\ \beta \in B. \tag{1}$$

Here q is a target class prior (with $|\log(q_k/q_1)| \le 8$) and β ranges over a small tempering grid $B = \{0.7, 1, 1.4\}$ that absorbs moderate miscalibration of g, in the spirit of bias-corrected calibration for label shift (Alexandari et al., 2020). H^C places *no restriction on the input process*: X_t may follow any covariate shift and may even be chosen adaptively from the past. It contains H^{glob} (with $\theta = (\pi, 1)$ when g is calibrated), every covariate shift that preserves the source posterior, and every label shift occurring at an unknown time. Its complement, a target posterior that is not repaired by any prior and tempering correction, is what we call *concept shift* in the action-typed sense; it includes concept changes in the classical sense and harmful covariate shifts under which the frozen posterior extrapolates poorly.

The monitor outputs an alarm time and an online active set $\mathcal{A}_t \subseteq \{X, L, C\}$ of witnesses

whose evidence has crossed. The reported action type is

$$\text{type}(\mathcal{A}_t) = \begin{cases} \text{concept (retrain/abstain)} & C \in \mathcal{A}_t, \\ \text{label (prior correction)} & C \notin \mathcal{A}_t,\ L \in \mathcal{A}_t, \\ \text{covariate (keep, monitor)} & \mathcal{A}_t = \{X\}, \\ \text{none} & \mathcal{A}_t = \varnothing. \end{cases}$$

The guarantees we seek are (i) control of false alarms under H^{glob}, both in the "ever alarm" sense and in the change-detection sense of average run length (ARL); and (ii) a *certified concept claim*: under H^C the probability of ever reporting "concept" is at most α_C, irrespective of covariate or label shift. We do not claim a guarantee for the covariate-versus-label distinction; Section 5 explains why and Section 7 measures it.

4 TSSD: witnesses and evidence

4.1 Paired witnesses for inputs and prevalence

The covariate and label witnesses are paired, source-referenced e-processes and are only summarised here (full formulas are in Appendix B). Each scores the current target and the current reference with the *same* $\mathcal{F}_{t-1}$-measurable function, clips each score to $[-1, 1]$ with a constant calibrated once on disjoint source-only streams, and forms the paired increment $D_t = \text{clip}((s_t - s'_t)/2, -1, 1)$. Under H^{glob} the target and reference are exchangeable given $\mathcal{F}_{t-1}$, so D_t is conditionally symmetric, bounded and mean zero.

Covariate witness X. The score is a windowed kernel mean-embedding contrast between the last 50 targets and references, from which the directions spanned by the classifier's class-ratio features $v(x) = (p_g(x)_j/\pi_j - 1)_{j<K}$ are projected out (with a frozen Gram matrix). The projection is meant to make the witness insensitive to pure prevalence changes, which also move the input marginal.

Label witness L. With a predictable class-prior estimate q_{t-1} obtained by Saerens–EM (Saerens et al., 2002) on the last 200 target cases, the score is $\log\{(q_{t-1}/\pi)^\top p_g(x)\}$ for unlabelled steps and $\log(q_{t-1,y}/\pi_y)$ for labelled steps. A *predictable gate* suspends the label bets while past covariate evidence is large (Section 4.3); skipped steps are identity updates.

4.2 Posterior-validity witness C

A concept witness could also be built as a paired, clipped score, in the same way as X and L. We report such a witness as an ablation ("paired C). Clipping caps the evidence that a single surprising label can carry, which makes it slow, and its null is only H^{glob}. TSSD replaces it by a likelihood-ratio e-process built on universal inference (Wasserman et al., 2020) that uses revealed labels only and tests the composite null H^C directly.

Alternative. Before Y_t is seen, the monitor forms a predictable predictive distribution $a_{t-1}(\cdot \mid X_t) = \sum_{j=1}^{3} w_{t-1,j}\, e_{t-1,j}(\cdot \mid X_t)$ that mixes three experts: (i) the label-shift plug-in $e_1 = \eta_{(q_{t-1},1)}$, which uses the EM prior estimate of the label witness; (ii) a confusion remap $e_2 = M_{t-1}^\top p_g(X_t)$, where M_{t-1} is a row-normalised, exponentially discounted (factor 0.85) matrix of soft counts between model classes and revealed labels, with prior mass $1/K$ per cell; it can represent label permutations and other conditional changes that no prior correction expresses; (iii) a refit expert $e_3 = \text{softmax}(W_{t-1}^\top \phi(X_t))$ with features $\phi(x) = (x, \log\max\{p_g(x), 10^{-4}\}, 1)$, fitted by ridge-penalised maximum likelihood on the 50 most recent revealed labels and shrunk towards the frozen logits ($\|W - W_0\|^2$ penalty, weight 1). The weights follow a fixed-share Bayesian mixture (Herbster and Warmuth, 1998): after each revealed label, $w_{t,j} \propto w_{t-1,j}\, e_{t-1,j}(Y_t \mid X_t)$, mixed with

Algorithm 1 One TSSD step (both modes share all scores).

Require: $\mathcal{F}_{t-1}$; target (X_t, R_t, Y_t); single-use reference (X'_t, Y'_t)

Compute covariate and label scores of target and reference from $\mathcal{F}_{t-1}$; form D_t^X, D_t^L
Read both label gates from evidence at $t-1$
Update X in test and SR mode; update L in each mode only if that mode's gate is open
if $R_t = 1$ **then**
    Form $a_{t-1}(\cdot \mid X_t)$; observe Y_t; update suffix suprema and $E^{(m)}$ for kept starts
end if
Record crossings; update the online active set and reported action type
Store X_t for the kernel windows; update q_t by windowed EM; if $R_t = 1$ update M_t

the uniform vector at rate 0.01 so that the mixture can switch after a change. Under the null the mixture loses at most $\log 3$ plus the switching cost relative to its best expert. Every component is $\mathcal{F}_{t-1}$-measurable, which is all that validity requires; the experts only determine power.

E-process. Let $i_1 < i_2 < \dots$ be the labelled times and $\ell_\theta(u{:}v) = \sum_{i_u \le i \le i_v} \log \eta_\theta(Y_i \mid X_i)$. For a start index m and current index n define the change-robust null maximum

$$S^\star(m{:}n) = \max\Big\{ \sup_\theta \ell_\theta(m{:}n),\ \max_{m<c\le n} \big[\sup_\theta \ell_\theta(m{:}c-1) + \sup_\theta \ell_\theta(c{:}n) \big] \Big\}$$

and

$$\log E_n^{(m)} = \sum_{u=m}^{n} \log a_{i_u - 1}(Y_{i_u} \mid X_{i_u}) - S^\star(m{:}n). \tag{2}$$

All suffix suprema $\sup_\theta \ell_\theta(c{:}n)$ are concave maximisations in $\log q$ for each $\beta \in B$ and are cached, so the split maximisation costs $O(n)$ additions per start. The test-mode concept evidence is $E^{(1)}$. Dividing by the change-robust maximum, rather than by the likelihood of a fixed θ, costs power: the numerator has to beat a null fitted with hindsight. This is the price of a claim that survives a label shift at an unknown time.

4.3 Two evidence modes

For a paired witness with bets Λ (covariate $\{0.5, 1\}$, label $\{0.01, 0.05, 0.1, 0.25, 0.5, 1\}$), each signed component $c = (s, \lambda)$ updates by $f_t^c = 1 + s\lambda D_t$.

Test mode uses the products $M_t^c = \prod_{i \le t} f_i^c$, averaged over components; a witness crosses when $\log E_t \ge \log(3/\alpha)$. It controls the probability of *ever* alarming under H^{glob}.

Shiryaev–Roberts (SR) mode uses the e-detector recursion (Shin et al., 2024)

$$R_t^c = f_t^c\,(R_{t-1}^c + 1), \qquad R_0^c = 0,$$

that is, $R_t^c = \sum_{j \le t} \prod_{i=j}^{t} f_i^c$, a sum of test martingales started at every past time. For the posterior-validity witness, $R_t^C = \sum_{j \le t} E_t^{(m(j))}$, where $m(j)$ is the first labelled index at or after j (starts after the last label contribute 1 each); to bound cost we keep the 40 most recent label-starts plus the first, and dropping summands can only decrease R_t^C. A witness crosses when $R_t \ge A$, with $A = 3m/\alpha$ for a design window m (we use m equal to the evaluation horizon: $m = 400$ for the Gaussian study). In SR mode the label gate is open when $\log(R_{t-1}^X/(t-1)) < 3$; in test mode when $\max_{s<t} \log E_s^X < 2$. Both gates are $\mathcal{F}_{t-1}$-measurable.

The SR mode removes a structural weakness of test-mode monitoring for change detection: a product martingale has negative log-drift under the null, so the wealth with which it meets a late change is small and the detection delay grows with the change time. The SR statistic restarts implicitly and has a delay that does not depend on how long the null regime lasted (Section 7.5).

5 Guarantees

Proofs are in Appendix A. Throughout, (A1) labels are revealed at random: R_t is independent of (Y_t, Y_t') given $(\mathcal{F}_{t-1}, X_t, X_t')$; (A2) every quantity used to score or gate step t, including clip constants and the label gate, is $\mathcal{F}_{t-1}$-measurable; (A3) for statements about the C witness under H^{glob}, the source posterior lies in the family (1), i.e. $P(Y \mid X) = \eta_{\theta_0}$ for some $\theta_0 \in \Theta$.

Theorem 1 (Global null, test mode)**.** *Under H^{glob} with (A1)–(A3) and a single-use reference drawn from P independently of the target,* $\Pr(\exists t : \max_k E_t^k \geq 3/\alpha) \leq \alpha$.

Theorem 2 (Global null, SR mode: average run length and finite window)**.** *Under the conditions of Theorem 1, let $\tau = \inf\{t : \max_k R_t^k \geq A\}$. Then $\mathbb{E}[\tau] \geq A/3$ and, for every $m \geq 1$, $\Pr(\tau \leq m) \leq 3m/A$. With $A = 3m/\alpha$ the TSSD SR alarm has $\Pr(\tau \leq m) \leq \alpha$ and $\mathbb{E}[\tau] \geq m/\alpha$.*

Theorem 3 (Certified concept claim)**.** *Assume (A1)–(A2) and H^C. The input process (X_t), the reference stream and the paired witnesses may behave arbitrarily. Then $\Pr(\exists t : E_t^{(1)} \geq 1/\alpha_C) \leq \alpha_C$; and for $\tau_C = \inf\{t : R_t^C \geq A_C\}$, $\mathbb{E}[\tau_C] \geq A_C$ and $\Pr(\tau_C \leq m) \leq m/A_C$.*

Proposition 1 (Finite source pools)**.** *Let the references be drawn uniformly* without replacement *from a pool of $N \geq T$ points sampled i.i.d. from P independently of the target stream. Then the reference sequence is i.i.d. P and independent of the targets, so Theorems 1–2 hold exactly up to horizon T. Drawing with replacement only guarantees validity for the empirical pool law.*

Two remarks delimit the guarantees. First, Theorem 3 is the typed guarantee: a reported "concept" state is a false claim only if the target posterior is prior-correctable, and that event has probability at most $\alpha_C = \alpha/3$ *even under arbitrary covariate shift and a label shift at an unknown time.* It requires (A1); under informative label reveal (e.g. labels obtained preferentially for errors) the claim is not protected. Second, no analogous guarantee is offered for "label" versus "covariate". A pure covariate shift generally changes the class prevalence (labels follow the source posterior on shifted inputs), so the hypothesis "no label shift" is not identifiable from the input marginal without a model of $P(X \mid Y)$. TSSD handles this distinction heuristically, through orthogonalisation and the predictable gate, and we measure its error rates rather than claim them.

6 Experimental protocol

All code is pure Python/NumPy/SciPy and is released with the paper. All configuration constants of TSSD were fixed before any reported stream was generated: windows, bet grids, gates, the tempering grid, the alternative mixture weight, discount factors and SR starts. Every evaluation seed (≥ 30000) is disjoint from the clip-calibration seeds (9000–9019). Nothing was tuned on evaluation streams. The 11 ablations in Section 7.6 report the sensitivity to each constant.

6.1 Data and shifts

Controlled Gaussian streams. $K = 3$ classes, $d = 10$, uniform prior, $X \mid Y = y \sim N(\mu_y, I)$, with a multinomial logistic model fitted on 2000 source pairs (well specified for this family). Streams have $T = 400$ steps, a change at step 201, label reveal rate $\rho = 0.05$, and a fresh source reference per step. The six alternatives are as follows. *Covariate*: every class cloud is translated by $s \cdot (2, \ldots, 2)$ and labels are drawn from the source Bayes posterior, so $P(Y \mid X)$ is unchanged (this also changes prevalence). *Label*: prior $(0.7, 0.2, 0.1)$ with $P(X \mid Y)$ unchanged. *Concept*: inputs from the source marginal and the observed class cycled by one. The mixtures *covariate+label*, *covariate+concept* and *label+concept* apply both mechanisms. Under covariate+label the target

posterior equals a prior correction of the source posterior with a modified prior, so it belongs to H^C and the correct action type is "covariate" or "label" but never "concept". Sweeps vary the covariate scale $s \in \{0.1, 0.2, 0.35, 0.5\}$, the label mixing $\gamma \in \{0.25, 0.5, 0.75\}$ (prior $(1-\gamma)\pi + \gamma(0.7, 0.2, 0.1)$), the fraction of cycled labels $\in \{0.1, 0.25, 0.5\}$, $\rho \in \{0.02, 0.05, 0.10, 0.20\}$ and the change time $t_0 \in \{100, 500, 1000, 2000\}$ (horizon $t_0 + 300$). There are 200 streams per cell (100 for change times), 1000 null streams for calibration and 1000 independent null streams for evaluation.

Digits (exact-validity protocol). The 1797 handwritten digits ($d = 64$, $K = 10$) are split once into a fitting set (719), a reference pool and a target pool (539 each). Every stream draws its references and its targets *without replacement* from their pools (Proposition 1). Streams have $T = 300$ steps with a change at step 151. Covariate shift is an acquisition corruption (contrast $\times 1.35$ and Gaussian noise $\sigma = 0.45$); label shift resamples to 75% mass on digits 0–2; concept shift cycles the digit label; the mixtures combine these. We use 300 calibration nulls, 300 evaluation nulls and 100 streams per shift.

Covertype (natural geographic shift). We fit the source model on 4000 cases of wilderness area 0 (Rawah) using the ten cartographic variables ($K = 7$). A further 20,000 area-0 cases form the reference pool, and the remaining area-0 cases are a held-out same-domain target pool, which gives exact null streams. Each shifted stream switches at step 201 (of 600) to the natural population of area 1, 2 or 3 (Neota, Comanche Peak, Cache la Poudre).

Electricity (chronological drift). The Elec2 market data (Harries, 1999) has 45,312 half-hourly records and an up/down price label. The model is fitted on the first 4000 records, and the next 4000 are split at random into a reference pool and a source-period null pool. We monitor 18 consecutive 1000-record segments (starting at records 8000, 10000, ..., 42000) in chronological order, with 20 independent reveal and reference draws per segment.

Offline oracle typing for natural shifts. For natural shifts the true mechanism is unknown, so we characterise each target domain or segment *offline with all of its labels*, which is information the monitor never sees. A random half of the target data is used to fit (i) the best prior-and-tempering correction from family (1) and (ii) a refitted logistic model. On the other half we compare their log losses with that of the frozen model, and we also compute a domain-classifier AUC for the inputs. The oracle type is "concept" when the refit beats the best prior correction by more than 0.05 nats with a bootstrap lower bound above zero. Otherwise it is "label" if the prior correction gains more than 0.02 nats over the frozen model, and "covariate" if the AUC exceeds 0.6.

6.2 Comparators

We re-implemented ten detectors and ran them on the identical streams.

- BBSD-KS on classifier outputs in 50-case blocks (Rabanser et al., 2019).
- The multiclass classifier-CUSUM of Evans and G'Sell (2024), using 168 alternative priors.
- Page–Hinkley on revealed log loss (Page, 1954).
- DDM (Gama et al., 2004) and ADWIN (Bifet and Gavaldà, 2007) on revealed errors.
- The risk-tracking betting test of Podkopaev and Ramdas (2022), for an increase in 0–1 risk.
- Kernel two-sample testing by betting (Shekhar and Ramdas, 2023), with an online-Newton-step bet on the paired MMD witness.
- The sequential classifier two-sample test of Jang et al. (2022), with an online logistic discriminator.

- A conformal test martingale on revealed log-loss scores (Vovk et al., 2021; Vovk, 2020).
- A WATCH-style covariate-weighted conformal test martingale (Prinster et al., 2025), with online density-ratio weights (monitoring component only, without adaptation).

The *specialist ensemble* raises an alarm when BBSD, classifier-CUSUM or Page–Hinkley fires, each at $\alpha/3$. It reports concept if Page–Hinkley fired, label if CUSUM fired, and covariate if only BBSD fired.

Methods with their own guarantee are reported at their native threshold (anytime-valid tests at $1/\alpha = 20$; DDM at its 3-sigma rule; ADWIN at $\delta = 0.002$). In addition, *every* method, TSSD included, is calibrated to a matched 5% horizon false-alarm rate on the 1000 calibration nulls and evaluated on the independent nulls and shifted streams. For TSSD the matched threshold is a common shift of the three log thresholds.

Metrics. A stream with a crossing before the change is a pre-change (false) alarm and is never counted as a detection. Detection delay is the crossing step minus the change step. The reported type is the online state at the horizon. For each condition we report detections, typed-correct detections (the type is in the condition's allowed set) and the median delay.

7 Results

7.1 False alarms and detection on controlled streams

Table 2 reports detection on the Gaussian study. At the theoretical thresholds, TSSD alarms on 1.5% (test mode) and 1.3% (SR mode) of the 1000 evaluation nulls. Both rates are well inside their guarantees ($\leq 5\%$ ever, and $\leq 5\%$ within 400 steps). Pre-change alarms on shifted streams occur at the same rate.

Two patterns stand out. First, every comparator is a specialist. Output and input detectors (BBSD, CUSUM, the kernel and classifier two-sample tests) find covariate and label shifts but essentially never concept shifts, because a label permutation leaves the input and output marginals unchanged. Label-based detectors (Page–Hinkley, DDM, ADWIN, risk tracking, the conformal and WATCH-style martingales) find concept shifts but essentially never covariate or label shifts. TSSD in SR mode is the only single method that detects all six alternatives at a high rate: 180–200 of 200 streams, at a 1.3% null alarm rate. Second, on the concept alternatives TSSD is slower than the best label-based specialists: its median delay is 98 steps against 15 for Page–Hinkley at a matched false-alarm rate. This is the price of the certified claim discussed in Section 8. The specialist ensemble detects as broadly as TSSD and faster, so the case for TSSD rests on the reliability of its types, which we examine next.

The SR mode dominates the test mode on every alternative, for example 198 against 177 label detections with a delay of 82 against 120. At matched false-alarm rates the gap persists. The posterior-validity witness detects 180 concept streams, against 130 for a clipped paired concept witness built like X and L (ablation), with a median delay of 98 against 143 steps.

7.2 Reliability of the reported type

Table 3 counts the streams that are both detected and reported with an action type from the correct set. The correct sets are: covariate for Cov; label for Lab; concept for Con, C+C and L+C; and covariate or label, but never concept, for C+L.

TSSD in SR mode types 199/200 covariate, 198/200 label and 199/200 covariate+label streams correctly, and reports concept on 180–189 of the three concept alternatives. Among streams it detects, the wrong types are almost entirely "covariate" for C+C (11 streams whose concept evidence had not yet crossed at the horizon) and "label" for L+C (13 streams).

Table 2: Gaussian study (200 streams per shift; 1000 evaluation nulls). Entries are post-change detections out of 200, with the median delay in parentheses; "FA%" is the null alarm rate within 400 steps. "Matched" thresholds are calibrated to a 5% horizon false-alarm rate on 1000 disjoint nulls; "native" uses each method's own guarantee. Shift abbreviations: Cov = covariate, Lab = label, Con = concept; C+L, C+C and L+C are the mixtures covariate+label, covariate+concept and label+concept.

Method	FA%	Cov	Lab	Con	C+L	C+C	L+C
TSSD (theory thresholds)							
TSSD, SR mode	1.3	199 (22)	198 (82)	180 (98)	199 (22)	200 (22)	200 (70)
TSSD, test mode	1.5	198 (39)	177 (120)	177 (99)	199 (40)	197 (38)	195 (87)
Matched 5% horizon false-alarm rate							
TSSD, SR mode	5.0	197 (18)	197 (67)	184 (93)	196 (18)	197 (17)	199 (57)
TSSD, test mode	5.7	192 (34)	178 (107)	174 (92)	191 (34)	193 (32)	187 (77)
Specialist ensemble	6.5	194 (34)	198 (28)	195 (17)	197 (18)	199 (19)	197 (15)
BBSD-KS (Rabanser et al., 2019)	5.2	191 (49)	200 (49)	4 (74)	194 (49)	199 (49)	199 (49)
Classifier-CUSUM (Evans and G'Sell, 2024)	7.2	192 (29)	197 (25)	9 (100)	196 (17)	197 (30)	194 (26)
Page–Hinkley (Page, 1954)	4.8	13 (127)	3 (122)	197 (15)	129 (62)	199 (18)	199 (16)
DDM (Gama et al., 2004)	4.1	22 (97)	4 (135)	188 (41)	64 (96)	180 (38)	187 (37)
ADWIN (Bifet and Gavaldà, 2007)	4.4	16 (96)	3 (133)	198 (37)	48 (102)	197 (42)	199 (39)
Risk tracking (Podkopaev and Ramdas, 2022)	3.4	15 (129)	4 (68)	199 (45)	55 (116)	198 (50)	197 (44)
Kernel 2ST betting (Shekhar and Ramdas, 2023)	5.6	197 (17)	192 (68)	6 (92)	197 (16)	195 (17)	193 (68)
Seq. classifier 2ST (Jang et al., 2022)	7.3	190 (19)	188 (98)	3 (74)	193 (17)	193 (20)	191 (90)
Conformal TM (Vovk et al., 2021)	4.7	7 (129)	0 (–)	196 (39)	32 (80)	191 (42)	195 (39)
WATCH-style WCTM (Prinster et al., 2025)	4.2	7 (129)	2 (130)	196 (42)	12 (82)	187 (58)	194 (39)
Native anytime-valid thresholds							
Risk tracking	0.0	0 (–)	0 (–)	134 (144)	0 (–)	122 (152)	132 (139)
Kernel 2ST betting	0.0	200 (49)	2 (188)	0 (–)	200 (46)	200 (48)	0 (–)
Seq. classifier 2ST	0.0	200 (32)	157 (150)	0 (–)	200 (31)	200 (33)	169 (148)
Conformal TM	1.2	1 (149)	0 (–)	199 (51)	19 (123)	197 (55)	199 (49)
WATCH-style WCTM	1.9	2 (110)	0 (–)	198 (46)	4 (94)	184 (67)	198 (48)

Table 3: Typed-correct detections out of 200 (Gaussian study). "False concept" counts streams with a concept claim at any time where the posterior is prior-correctable: 1000 evaluation nulls plus the Cov, Lab and C+L arms, 1600 streams in all.

Method	Cov	Lab	Con	C+L	C+C	L+C	False concept
TSSD, SR mode	199	198	180	199	189	187	0
TSSD, test mode	198	177	177	199	185	182	0
TSSD, test mode (matched)	191	177	174	191	181	177	–
Specialist ensemble (matched)	0	195	195	87	199	197	149

The specialist ensemble detects broadly but types badly. None of the 194 detected covariate shifts is typed correctly: 192 are reported as label shifts, because the covariate shift changes prevalence and the label-shift CUSUM fires first, and 2 as concept shifts. Under covariate+label shift, 110 of 197 detected streams are reported as concept, which would trigger an unnecessary retraining in more than half of the deployments. In total, the ensemble's concept detector fires on 149 of the 1600 streams whose posterior is prior-correctable (30 of them nulls).

The *certified* concept claim made no false claims anywhere: it did not fire on any of the 2000 null streams, on any covariate, label or covariate+label stream, or (Section 7.7) on any real-data stream whose posterior is prior-correctable. This holds in both modes.

7.3 Shift magnitude

Figure 1 varies the strength of each shift. Thresholds are matched for the comparators and at the theoretical value for TSSD; each curve is the fraction of 200 streams detected (for TSSD and the ensemble, detected *and* correctly typed).

Covariate. TSSD is the most sensitive covariate detector in the study. At one tenth of the base translation it detects and correctly types 114/200 streams, against 44 for the best specialist (kernel two-sample betting) and 3 for the ensemble. At 0.2 it types 200/200. The ensemble's typed

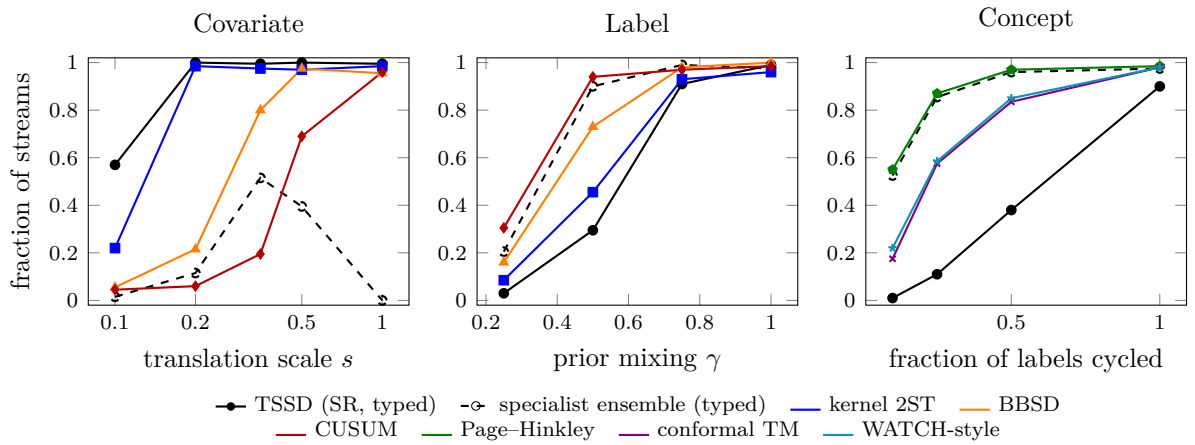


Figure 1: Detection versus shift magnitude (Gaussian, 200 streams per point; level 1 is the main study). TSSD and the ensemble count typed-correct detections; the other detectors count detections. Comparators use matched 5% thresholds; TSSD uses its theoretical SR threshold.

accuracy stays poor at every magnitude (79/200 at half strength), because prevalence-driven CUSUM alarms label most covariate shifts as label shifts.

Label. Mild prevalence changes are TSSD's weakest regime. At $\gamma = 0.5$ the classifier-CUSUM detects 188/200 and TSSD 59/200, and only at $\gamma = 0.75$ does TSSD reach 182/200. The label witness has to work through the gate and the orthogonalised covariate witness, while the CUSUM is optimised for exactly this alternative.

Partial concept. When only a fraction of labels is cycled, the certified concept claim needs much more evidence than an uncorrected loss monitor. With half of the labels cycled, TSSD certifies 76/200 streams while Page–Hinkley detects 194/200; at 10% cycling TSSD certifies essentially none. This is the clearest instance of the price of certification. A partial label permutation lies close to the prior-correctable family, and with about ten post-change labels the null fitted with hindsight absorbs most of the evidence.

7.4 Label budget

Figure 2(a) and Table 4 vary the reveal rate; Figure 2(b) shows the three test-mode evidence processes on one label+concept stream. The covariate and label witnesses hardly depend on it. The certified concept claim depends on it strongly: it types 70/200 concept streams correctly at 2%, 180/200 at 5%, 198/200 at 10% and 199/200 at 20%, where the median delay drops to 22 steps. From 10% upwards it matches the conformal test martingale's detection rate, while also certifying the type. At 2% the clipped paired witness of the ablation detects only 5/200. Across all four rates, no concept claim was made on any of the 4400 streams whose posterior is prior-correctable.

7.5 Change time: test mode versus SR mode

Table 5 moves the change point from step 100 to step 2000 (100 streams per cell, horizon $t_0 + 300$). The test-mode delay grows with the length of the null period, from 35 to 147 steps for covariate shift and from 108 to 186 for label shift. The test martingale's wealth decays under the null, so a late change finds it far below its starting point. The SR delay is flat: about 23 steps for covariate shift and about 75 for label shift, whatever the change time.

The concept witness is almost unaffected in either mode. Before the change its alternative tracks the null closely, so its evidence barely decays.

The price of restarting is a higher alarm rate over long null stretches. On 200 null streams of 2300 steps, SR mode raises 23 alarms and test mode 8. Both are consistent with their guarantees:

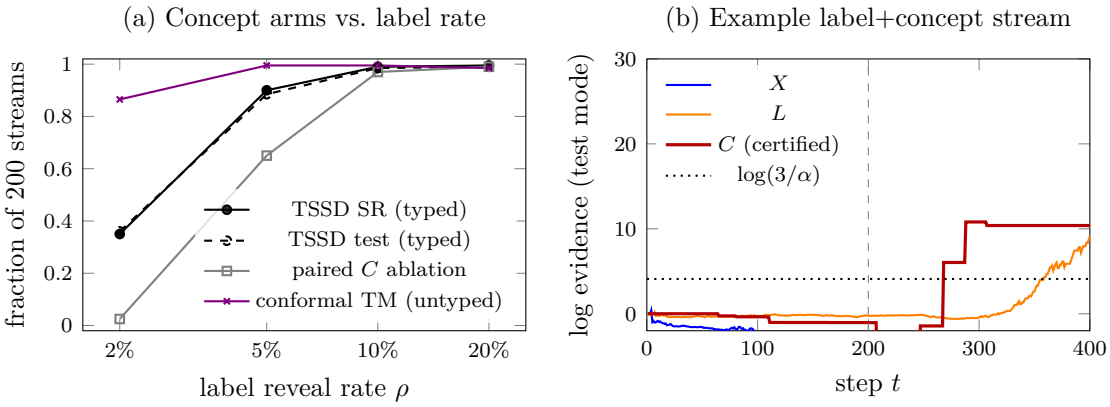


Figure 2: (a) Typed-correct concept detections versus label budget (Gaussian). (b) Test-mode log evidence on one label+concept stream; the change is at step 201 (dashed). The certified concept witness crosses about 70 steps after the change and fixes the reported action type as concept; the label witness crosses only near the horizon, and the covariate witness stays flat.

an ARL of at least $m/\alpha = 8000$ steps for SR, and at most 5% ever for test mode. Within the 400-step design window each mode raises 3. These pre-change alarms are the main reason SR detection falls from 99–100 to 90–94 streams at $t_0 = 2000$. For long deployments, A can be set from the desired ARL rather than from a fixed window. Across all 1000 change-time streams with a prior-correctable posterior, no concept claim was made.

7.6 Ablations

Table 6 perturbs each design constant on fresh streams: 300 nulls and 100 streams per shift. The default configuration corresponds to the main study (Table 3).

No perturbation produced a false concept claim. Null alarm rates stayed at or below 4/300. The covariate, label and covariate+label results move by at most a few streams in SR mode, and the test-mode label witness is sensitive only to a tighter gate (65 versus 86 typed-correct).

The concept results reveal a robustness–power trade-off in the null family. Removing tempering ($B = \{1\}$) or narrowing the prior box ($|u_k| \leq 4$) raises the typed concept rate from 87 to 97–99. A wider tempering grid lowers it to 76. On the well-specified Gaussian model the smaller family is safe. We nevertheless keep the tempered family as the default, because it is what protects the concept claim against moderate miscalibration on real data, where the source model is not well specified. The two-component alternative without the refit expert performs similarly on Gaussian data (85 versus 87); the refit expert matters most when the posterior change involves directions that the frozen logits do not represent.

7.7 Real data

Digits. Table 7 reports the exact-validity Digits study. The null alarm rates are 1/300 (SR) and 3/300 (test) at 5% labels, and 2/100 and 3/100 at 20%. The acquisition corruption is detected and typed as covariate in all streams. Prevalence change is TSSD's weak spot here: SR mode types 53/100 label streams correctly against 99/100 detections by classifier-CUSUM. The test mode is weaker still, because the covariate gate often closes before the label evidence accumulates. With ten classes and about seven post-change labels at 5%, the certified concept claim fires on only 12–31% of the concept alternatives. At 20% labels it fires on 100/100 in each of the three concept arms, and it makes no false claim on any null, covariate, label or covariate+label stream at either rate. Unlike on the Gaussian data, the specialist ensemble types Digits well at 5% labels: CUSUM is silent under the acquisition corruption, so the ensemble's characteristic failure does

Table 4: Label-budget sensitivity (Gaussian, 200 streams per cell; theory thresholds). Entries are typed-correct detections; "null" is the number of null streams with any alarm. CTM = conformal test martingale (native threshold), counted as detections.

Rate	Condition	TSSD SR	TSSD test	Paired C (ablation)	CTM
2%	null (any alarm)	0/200	6/200	6/200	1/200
	Cov	200/200	197/200	197/200	0/200
	Lab	199/200	181/200	181/200	0/200
	Con	70/200	72/200	5/200	173/200
	L+C	84/200	81/200	7/200	169/200
5%	null (any alarm)	13/1000	15/1000	15/1000	12/1000
	Cov	199/200	198/200	198/200	1/200
	Lab	198/200	177/200	177/200	0/200
	Con	180/200	177/200	130/200	199/200
	L+C	187/200	182/200	137/200	199/200
10%	null (any alarm)	3/200	8/200	8/200	2/200
	Cov	200/200	199/200	199/200	1/200
	Lab	199/200	178/200	178/200	1/200
	Con	198/200	197/200	194/200	199/200
	L+C	200/200	197/200	193/200	197/200
20%	null (any alarm)	2/200	2/200	2/200	0/200
	Cov	199/200	199/200	196/200	9/200
	Lab	200/200	172/200	171/200	0/200
	Con	199/200	198/200	198/200	197/200
	L+C	200/200	198/200	197/200	199/200

Table 5: Change time (Gaussian, 100 streams per cell). Typed-correct detections with median delay in parentheses, test mode versus SR mode.

	Covariate		Label		Concept	
t_0	test	SR	test	SR	test	SR
100	98 (35)	99 (23)	91 (108)	100 (90)	99 (98)	99 (99)
500	98 (61)	100 (22)	82 (126)	99 (74)	98 (96)	98 (94)
1000	97 (85)	96 (22)	93 (153)	99 (75)	100 (107)	99 (94)
2000	94 (147)	90 (23)	89 (186)	94 (76)	99 (101)	93 (95)

not arise. Its concept reports remain uncertified, however[1].

Covertype: natural geographic shift. On exact same-domain nulls (held-out Rawah data), TSSD alarms on 3/300 (SR) and 2/300 (test) streams. The ensemble, calibrated on disjoint nulls of the same kind, alarms on 27/300. All three natural target domains are detected by TSSD in 96–100% of streams (median SR delay 31–57 steps). They are reported as covariate or label shifts, and the certified concept claim is never made at 5% labels and is made in 3–4 of 30 streams at 20%.

The offline oracle (Table 8) explains this. In every domain the refitted model beats the best prior correction by 0.19–0.37 nats per case, so the oracle type is concept. But a prior correction already removes 43%, 75% and 98% of the frozen model's excess loss in areas 1, 2 and 3. Certifying the remaining gap of a few tenths of a nat per label, against a null fitted with hindsight over seven classes, needs more labels than the 20–80 available. The ensemble reports concept on most area-2 and area-3 streams. This agrees with the oracle, but the ensemble's concept claims are uncertified: the same rule produced 149 false concept claims on the Gaussian streams. We read Covertype as

[1]The ensemble's thresholds are matched at 5% labels. At 20% its null alarm rate rises to 32/100, so its 20% entries are not shown.

Table 6: Ablations (Gaussian; 300 nulls, 100 streams per shift). Each cell gives SR / test-mode values: null streams with any alarm, or typed-correct detections. The last column counts concept claims (either mode) on the 700 streams whose posterior is prior-correctable.

Perturbation	Null	Cov	Lab	Con	C+L	L+C	False C
kernel window 25	1 / 4	99 / 97	100 / 89	87 / 87	100 / 100	90 / 90	0
kernel window 100	1 / 3	100 / 97	100 / 90	87 / 87	100 / 100	90 / 90	0
confusion discount 0.70	0 / 4	100 / 97	100 / 86	90 / 88	100 / 99	90 / 90	0
confusion discount 0.95	0 / 4	100 / 97	100 / 86	84 / 85	100 / 99	90 / 89	0
gates 1 / 2 (test / SR)	0 / 4	100 / 97	100 / 65	87 / 87	100 / 99	90 / 90	0
gates 3 / 4 (test / SR)	0 / 4	100 / 97	100 / 90	87 / 87	100 / 99	90 / 90	0
tempering grid $B = \{1\}$	0 / 4	100 / 97	100 / 86	99 / 99	100 / 99	98 / 98	0
$B = \{0.5, 0.7, 1, 1.4, 2\}$	0 / 4	100 / 97	100 / 86	76 / 78	100 / 99	72 / 69	0
EM on full history	0 / 4	100 / 97	99 / 83	87 / 87	100 / 99	90 / 90	0
basic two-component alternative	0 / 4	100 / 97	100 / 86	85 / 89	100 / 99	90 / 89	0
null box $\lvert u_k \rvert \leq 4$	0 / 4	100 / 97	100 / 86	97 / 96	100 / 99	97 / 96	0

Table 7: Digits (exact-validity protocol, 100 streams per shift). Entries are typed-correct detections for TSSD and the ensemble and detections for the other methods. Comparators use matched thresholds except CTM and WATCH-style, which use their native anytime-valid thresholds.

	Cov	Lab	Con	C+L	C+C	L+C
5% labels						
TSSD, SR mode	100	53	12	100	31	15
TSSD, test mode	100	4	12	99	29	15
Specialist ensemble	94	98	98	96	97	99
Classifier-CUSUM	3	99	1	97	5	97
BBSD-KS	98	19	6	99	93	16
Kernel 2ST betting	100	18	4	100	99	12
Page–Hinkley	11	5	98	12	99	100
Conformal TM (native)	0	0	96	0	98	98
WATCH-style (native)	0	0	100	0	97	98
20% labels (theory/native thresholds)						
TSSD, SR mode	100	44	100	100	100	100
TSSD, test mode	99	3	98	99	99	98
Conformal TM (native)	0	0	100	1	99	100
WATCH-style (native)	0	0	99	1	98	99

an honest limitation. TSSD's "covariate" state means *inputs moved and no posterior change is certified*, not *the model is safe*. When harm matters more than type, it should be combined with a risk monitor (Section 8).

Electricity: chronological drift. Table 9 monitors 18 consecutive segments of the Elec2 market data in time order.

On exchangeable source-period nulls, TSSD alarms on 2/300 (SR) and 4/300 (test) streams. All comparators were calibrated to 5% on these nulls.

The offline oracle splits the segments into two regimes. In the first five (records 8000–17999) the input distribution has moved (AUC 0.55–0.79) and a prior correction removes most of the excess loss (gap ≤ 0.053 nats). In the remaining 13 (records 18000 onwards) the frozen posterior fails badly (frozen log loss 5.7–15.3) and only a refit repairs it (gap 2.2–8.8 nats).

TSSD follows this structure exactly. In the first regime it raises an alarm in every run and reports a covariate shift, never a concept shift. In the second regime it makes a *certified* concept claim in all 260 runs (median SR delay 24 records). The loss-based detectors agree on the second regime: Page–Hinkley fires in all of its runs, and the conformal martingale in 15–20 of 20 per segment. The WATCH-style weighted martingale fires in only 4–9 of 20 runs on four of these segments. Its covariate weighting absorbs part of the change when the inputs move strongly, the behaviour it is designed to have for benign covariate shift. Among the input detectors, BBSD fires

Table 8: Offline full-label characterisation of the natural shifts (not available to any monitor). LL: log loss on held-out target data (nats per case); prior: best prior and tempering correction in the family (1); refit: logistic model refitted on target data; AUC: input domain classifier.

Domain / segment	acc. frozen	LL frozen	LL prior	LL refit	gap	AUC	oracle
Covertype area 0 (held-out)	0.77	0.56	0.56	0.56	-0.00	0.49	none
Covertype area 1	0.63	0.95	0.81	0.62	0.19	0.98	concept
Covertype area 2	0.63	2.22	1.11	0.74	0.37	0.81	concept
Covertype area 3	0.08	18.67	1.17	0.82	0.35	1.00	concept
Elec2 source period	0.75	0.46	0.38	0.37	0.01	0.52	label
Elec2 segment 8000	0.84	0.38	0.28	0.23	0.05	0.79	label
Elec2 segment 12000	0.76	0.46	0.41	0.38	0.03	0.77	label
Elec2 segment 16000	0.70	0.59	0.45	0.41	0.04	0.59	label
Elec2 segment 18000	0.74	7.03	4.54	0.20	4.34	0.95	concept
Elec2 segment 24000	0.64	9.78	7.03	0.51	6.52	0.60	concept
Elec2 segment 34000	0.44	15.32	9.43	0.66	8.77	1.00	concept

Table 9: Electricity: chronological segments of 1000 records (20 runs each, 5% labels). The oracle type and gap come from Table 8's procedure. TSSD columns give alarm counts and the SR action-type counts concept/label/covariate. Comparator columns give alarm counts at matched thresholds (PH = Page–Hinkley; CTM = conformal TM; WATCH = WATCH-style WCTM).

Segment start	Oracle	Gap	SR alarm	Test alarm	C / L / X	PH	CTM	WATCH	BBSD
Source period (null)	–	–	2/300	4/300	0 / 0 / –	14	14	14	13
8000	label	0.05	20/20	20/20	0 / 0 / 20	0	0	0	20
10000	concept	0.05	20/20	20/20	0 / 0 / 20	0	0	0	20
12000	label	0.03	20/20	20/20	0 / 0 / 20	0	0	0	20
14000	label	0.05	20/20	20/20	0 / 0 / 20	3	0	0	20
16000	label	0.04	20/20	20/20	0 / 0 / 20	2	1	0	20
18000	concept	4.34	20/20	20/20	20 / 0 / 0	20	17	18	20
20000	concept	8.09	20/20	20/20	20 / 0 / 0	20	20	20	20
22000	concept	6.42	20/20	20/20	20 / 0 / 0	20	19	9	20
24000	concept	6.52	20/20	20/20	20 / 0 / 0	20	20	20	20
26000	concept	2.22	20/20	20/20	20 / 0 / 0	20	15	18	20
28000	concept	4.25	20/20	20/20	20 / 0 / 0	20	19	20	20
30000	concept	4.21	20/20	20/20	20 / 0 / 0	20	19	19	20
32000	concept	4.14	20/20	20/20	20 / 0 / 0	20	19	4	20
34000	concept	8.77	20/20	20/20	20 / 0 / 0	20	20	20	20
36000	concept	8.53	20/20	20/20	20 / 0 / 0	20	20	4	20
38000	concept	7.31	20/20	20/20	20 / 0 / 0	20	19	5	20
40000	concept	6.81	20/20	20/20	20 / 0 / 0	20	20	20	20
42000	concept	4.54	20/20	20/20	20 / 0 / 0	20	19	18	20

in both regimes. TSSD is the only method whose output distinguishes them with a guarantee.

The early segments use labels that are serially dependent and only roughly exchangeable, so the absence of concept claims there is descriptive rather than a test of validity.

8 Discussion and limitations

What the typed guarantee buys. The specialist ensemble is the natural engineering baseline, and on raw detection it is competitive or better: Page–Hinkley and the conformal martingales detect concept changes sooner than any TSSD witness. Its *types*, however, are not calibrated. A covariate shift that changes prevalence fires the label-shift CUSUM, and a mixed covariate–label shift fires the loss monitor and is reported as concept. Each of these errors maps to a costly action: an unnecessary prior correction, or an unnecessary retraining. TSSD's concept claim cannot be fooled in this way at more than its nominal rate, whatever the inputs do. Across all our experiments, on 16,830 streams with or without covariate or label shift, the claim never fired on a stream whose target posterior was prior-correctable.

The price. The certified claim pays twice. First, the alternative must beat a null fitted with hindsight that also allows a change point, so concept evidence accumulates more slowly than in an uncorrected loss monitor. Second, the evidence is in labels: with $K = 10$ classes and about seven post-change labels (Digits at 5%), or with natural shifts in which a prior correction already removes most of the excess loss (Covertype), a certified claim is out of reach at a 5% label budget, and TSSD correctly does not make it. With four times the label budget the Digits concept claims become reliable. This information bottleneck is intrinsic to the claim, not an artefact of the implementation: no procedure can certify "not prior-correctable" faster than the labels reveal it.

Covariate versus label. The distinction between these two types remains heuristic. A covariate shift whose labels follow the source posterior generally changes prevalence, so "no label shift" is not identifiable from inputs alone without a model of $P(X \mid Y)$. TSSD's orthogonalisation and predictable gate make the distinction work in our controlled and natural settings, but the error rates in Section 7.2 are empirical. The action consequence of this residual error is mild, since both types leave the posterior prior-correctable.

Assumptions. The guarantees require random label reveal (A1). Under active or error-driven labelling the concept claim is not protected. The global-null statements for the C witness also require the source posterior to lie in the tempered prior-correction family (A3). The tempering grid gives some slack, and our null and same-domain streams showed no false concept claims, but a badly miscalibrated source model would need recalibration before deployment. Time-series dependence, as in Electricity, violates the exchangeability that underlies the paired witnesses. Those segments are therefore descriptive rather than tests of validity, and the null-period check there is only indicative. Finally, H^C allows one change of the prior-correction parameter per SR start. A drifting prior is covered only approximately, through the SR restarts.

Cost. A 400-step stream takes 1–2 s on one CPU core in pure Python. The cost of the C witness grows quadratically in the number of revealed labels within an SR start, which we cap at 40 starts. For high label rates or very long streams, windowing the suffix computation, or replacing it with a sufficient-statistic update, is an obvious engineering improvement.

9 Conclusion

TSSD turns three sources of evidence (unlabelled inputs, prevalence and sparse labels) into a single monitor with an anytime-valid alarm, a restarting change-detection mode with average-run-length control, and a concept claim certified against covariate and label shift. Across controlled, semi-real and naturally shifted data, and against ten detectors from the recent literature, the experiments support a precise claim. TSSD is not the fastest detector of each shift type, but it is the only one among them whose typed output is reliable enough to drive the keep/recalibrate/retrain decision. Natural next steps are an analogous guarantee for the covariate/label distinction under explicit class-conditional models, and label-efficient alternatives for large label spaces.

Statements and declarations

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Data availability. Digits is distributed with scikit-learn (Alpaydin and Kaynak, 1998), Covertype is available from the UCI repository (Blackard and Dean, 1999), and Electricity (Elec2) from

OpenML (Harries, 1999). The Gaussian streams are generated by the released code. All derived results are included in the repository.

Code availability. All code (pure Python/NumPy/SciPy), raw per-stream summaries and the scripts that regenerate every table and figure are released with the paper.

Use of AI tools. AI assistants were used for code development, consistency checking and editing. The authors are responsible for verifying all code, results, citations and text.

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A Proofs

Lemma A (paired increments). Let $\mathcal{G}_t = \sigma(\mathcal{F}_{t-1}, R_t)$. Under H^{glob} with a single-use reference drawn from P independently of the target, (Z_t, Z_t') is exchangeable given $\mathcal{G}_t$ by (A1). The paired witnesses apply one $\mathcal{G}_t$-measurable score s to both cases, so swapping them maps D_t to $-D_t$ (the outer clip is odd). Hence D_t is conditionally symmetric, $|D_t| \le 1$ and $\mathbb{E}[D_t \mid \mathcal{G}_t] = 0$. For $\lambda \in [0,1]$ and $s \in \{\pm 1\}$, $f_t = 1 + s\lambda D_t \ge 0$ and $\mathbb{E}[f_t \mid \mathcal{G}_t] = 1$. A predictably skipped update ($f_t \equiv 1$) has the same property.

Lemma B (likelihood-ratio factors). Let θ_t^* be the true parameter path under H^C and $\mathcal{H}_t = \sigma(\mathcal{F}_{t-1}, X_t, R_t)$. On a labelled step, $a_{t-1}(\cdot \mid X_t)$ is $\mathcal{H}_t$-measurable and sums to one, and by (A1) $Y_t \mid \mathcal{H}_t \sim \eta_{\theta_t^*}(\cdot \mid X_t)$. Therefore $\mathbb{E}[a_{t-1}(Y_t \mid X_t)/\eta_{\theta_t^*}(Y_t \mid X_t) \mid \mathcal{H}_t] = \sum_y a_{t-1}(y \mid X_t) = 1$. Unlabelled steps contribute the factor 1. Consequently, for every start j, $L_t^{(j)} = \prod_{\text{labelled } i \in [j,t]} a_{i-1}(Y_i \mid X_i)/\eta_{\theta_i^*}(Y_i \mid X_i)$ is a nonnegative martingale with $L_{j-1}^{(j)} = 1$, whatever the law of the inputs.

Lemma C (domination). The path θ^* restricted to the labelled indices $m, \dots, n$ is constant or has one change at some index c. In the first case $\sum_u \log \eta_{\theta^*}(Y_{i_u} \mid X_{i_u}) \le \sup_\theta \ell_\theta(m{:}n)$; in the second it is at most $\sup_\theta \ell_\theta(m{:}c-1) + \sup_\theta \ell_\theta(c{:}n)$. Either way it is at most $S^\star(m{:}n)$, so $E_n^{(m)} \le L_{i_n}^{(i_m)}$.

Proof of Theorem 1. By Lemma A, each component product of X and L is a nonnegative martingale started at one, and so is each finite average. By (A3) the global null is contained in H^C with a constant path, so Lemmas B–C give $E_t^C \le L_t^{(1)}$. Ville's inequality bounds each witness's crossing probability by $\alpha/3$; a union bound over the three witnesses gives the claim. No independence between witnesses is needed. □

Proof of Theorem 2. For a paired component, $\mathbb{E}[R_t^c \mid \mathcal{F}_{t-1}] = \mathbb{E}[f_t^c \mid \mathcal{F}_{t-1}](R_{t-1}^c + 1) = R_{t-1}^c + 1$, so $R_t^c - t$ is a martingale and R^c is a nonnegative submartingale with $\mathbb{E}R_t^c = t$; averages over components inherit both properties. For the posterior-validity witness, Lemmas B–C give $R_t^C \le \tilde{R}_t^C := \sum_{j \le t} L_t^{(j)}$, and the same computation shows $\tilde{R}_t^C - t$ is a martingale. Put $\tilde{R}^X = R^X$, $\tilde{R}^L = R^L$ and $S_t = \sum_k \tilde{R}_t^k$, so $S_t - 3t$ is a martingale and $S_t \ge \max_k R_t^k$. *Finite window.* $\Pr(\tau \le m) \le \sum_k \Pr(\max_{t \le m} \tilde{R}_t^k \ge A) \le \sum_k \mathbb{E}[\tilde{R}_m^k]/A = 3m/A$ by Doob's maximal inequality. *ARL.* If $\mathbb{E}\tau = \infty$ there is nothing to prove. Otherwise optional stopping at $\tau \wedge n$ gives $\mathbb{E}[S_{\tau \wedge n}] = 3\mathbb{E}[\tau \wedge n] \le 3\mathbb{E}[\tau]$, and Fatou's lemma gives $\mathbb{E}[S_\tau] \le 3\mathbb{E}[\tau]$. Since $S_\tau \ge A$ on $\{\tau < \infty\}$, $A \le 3\mathbb{E}[\tau]$. □

Proof of Theorem 3. Under H^C, Lemmas B–C hold with no assumption on the inputs, the references or the other witnesses. The test-mode claim follows from Ville's inequality applied to $L^{(1)} \ge E^{(1)}$. The SR claims follow from the argument of Theorem 2 with a single witness:

$R_t^C \le \tilde{R}_t^C$, $\tilde{R}_t^C - t$ is a martingale, $\Pr(\tau_C \le m) \le m/A_C$ and $\mathbb{E}[\tau_C] \ge A_C$. Discarding SR starts removes nonnegative summands and preserves every bound. □

Proof of Proposition 1. A uniformly random ordering of an i.i.d. sample is again i.i.d., and drawing without replacement reads the pool in such an order. The first T references are therefore i.i.d. P, and they are independent of the target stream because the pool is. Each reference is used once, so Lemma A applies at every step. □

B Witness details

Covariate witness. With W_t and W_t' the last ≤ 50 targets and references,

$$h_t(x) = \frac{1}{|W_t|} \sum_{u \in W_t} k(x,u) - \frac{1}{|W_t'|} \sum_{u \in W_t'} k(x,u),$$

with a Gaussian kernel of median-heuristic bandwidth on the source sample. The orthogonalised score is $h_t^{\perp}(x) = h_t(x) - c_t^{\top} G^{-1} v(x) - b_t$, where $G = n_0^{-1} V_0^{\top} V_0 + 10^{-8} I$ is frozen on the source sample (at most 2000 points), and c_t and b_t are the window differences of the source kernel means weighted by v and unweighted, respectively. Both depend only on the windows. For $t = 1$ both scores are zero.

Label witness. q_{t-1} is obtained by ten EM steps on the last 200 target cases, with revealed labels entering as counts, and is projected to $q_k \ge 10^{-3}$. Scores are clipped at the source-calibrated 95th percentile.

Calibration. Clip constants are the 95th percentiles of the absolute raw scores on 20 (Gaussian) or 10 (other data) disjoint source-only streams. They are fixed before any monitored stream is evaluated. The posterior-validity witness has no clip.

Complexity. A step costs $O(d(w + n_0))$ for the covariate witness and $O(wK)$ for EM. On a labelled step with n labels so far, the C witness solves $n|B|$ warm-started concave programs in $K - 1$ variables. A 400-step Gaussian stream takes 1–2 s in pure Python/NumPy on one CPU core.